StrumStart: A Dataset of 1-on-1 Guitar Lessons for Speech-Music Co-reasoning

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Abstract

Recent attempts to integrate speech, music, and audio processing into Large Audio Language Models (LALMs) have relied on the combination of large datasets which were typically designed or collected for a particular audio sub-domain. In these datasets, it is often difficult to find data that interacts with multiple domains simultaneously. To help introduce datasets that specifically target co-reasoning abilities across speech, music, and audio, we focus on collaborative-music production as a natural speech-music co-reasoning environment and introduce StrumStart, a collection of 4 hours of 1-on-1 guitar lessons and a preliminary set of question-answering pairs. Furthermore, we evaluate how five Large Audio Language Models (LALMs) handle speech-music co-reasoning questions derived from the StrumStart dataset, demonstrating that model performance decreases on questions that specifically target Speech-Music Co-reasoning.

Index Terms: speech recognition, human-computer interaction, computational paralinguistics

1. Introduction

With the recent high-level of performance of Large Audio Language Models (LALMs) across a wide-range of speech, music, and audio tasks [1, 2], there has been an explosion of interest in understanding and improving the performance of LALMs on complex open-ended tasks [3, 4]. While LALMs perform individual audio, speech, and music tasks well, difficulties arise on tasks where more complex reasoning is required, such as deduction [5, 6].

One such task, Speech-Audio Co-reasoning (SAC), requires LALMs to combine information and infer relationships from both speech and audio events to correctly answer open-ended questions. As SAC requires a base-level of modeling of audio and speech task separately, many publicly available speech or audio datasets do not have SAC tasks explicitly provided, with recent work generating synthetic examples or using privately held data to explore such questions [6, 1]. There is a current lack of publicly available data collected in naturalistic environments specifically for the purpose of SAC.

To help address this gap, we create the StrumStart dataset, a collection of six 30 minute to 1 hour 1-on-1 guitar lessons alongside an initial set of 580 question-answer (QA) pairs that specifically aim to target Speech-Music Co-reasoning (SMC). As hinted at by the name Speech-Music Co-reasoning, guitar lessons, and 1-on-1 music lessons more broadly, serve as a natural training/testing ground for SMC capabilities. Seen in Figure 1, to perform SMC well, a LALM must be able to understand music and speech events jointly, determining the relationships between various speech and music events and drawing valid inferences from these relationships and events.

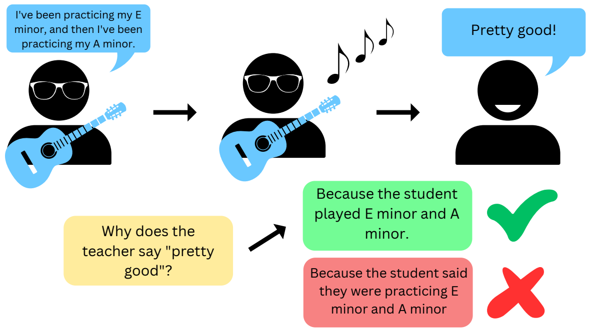

Figure 1: *Diagram of example Speech-Music Co-reasoning question from StrumStart. Misunderstandings of the relationship between speech and music events results in incorrect inferences.*

Furthermore, we assess the performance of five LALMs on StrumStart, LTU, LTU-AS, GAMA, Qwen, and SALMONN, [7, 2, 6, 8, 1], exploring how performance of these models vary across training data and tasks (Speech-Only Reasoning vs. Speech-Music Co-reasoning vs. Music-Only Reasoning). We demonstrate that, despite the better performance of SALMONN and LTU-AS on the QA tasks in comparison to the other models, that both models struggle in Speech-Music Coreasoning tasks and Music-Only tasks.

Our contributions are summarized as follows:

- StrumStart[1], a dataset of six 30min to 1hr 1-on-1 guitar lessons, along with transcripts and associated 580 handcrafted question-answer pairs for Speech-Only reasoning, Speech-Music Co-reasoning, and Music-Only reasoning tasks.
- An evaluation of five existing LALMs, LTU, LTU-AS, GAMA, Qwen, and SALMONN, on StrumStart, which demonstrates an explicit decreased performance on SMC tasks.

2. Related Work

While model architecture and prompting heavily influence how LALMs process multimodal information [4, 3], clever meta-dataset construction is crucial to the multi-modal capabilities of LALMs, with many research teams generating datasets for their

[1] Samples are available at https://SpeechMusicLessonsIS2025.github.io/. Full dataset release upon publicbation.

own models [9, 6, 1, 2, 10, 11, 12, 8]. Many of these datasets follow similar patterns of construction: gather task-specific or large publicly available datasets of speech, music, and audio, and use pre-trained LLMs like ChatGPT to generate question-answer pairs for LALM training/evaluation [13, 14]. [5] notes that many models trained using this sort of approach end up falling short on more complex open-ended tasks, and they propose a similar method of synthetic data generation via the notion of entailment to specifically test these models' deductive capabilities.

Our work primarily concerns one such difficult task, Speech-Audio Co-reasoning. [1] first proposes SAC, defining it as one of the most difficult multimodal tasks, reuqiring models to answer queries using audio, music, or speech evidence from a given clip. Sadly, the data used in [1] is closed-source and the dataset construction is unknown.

Fortunately, [6] tackles SAC under the broader banner of Complex Audio Reasoning, which generalizes SAC to understanding every individual acoustic event and their relationships, and being able to draw corresponding inferences. Towards this end, [6] release the CompA-R corpus, which generates complex instruction-response pairs by automatically extracting and combining information from audio and video.

While extensions to typical synthetic data generation for LALM training has been shown to improve co-reasoning capabilities of these models, it is still unclear what environments and interactions could serve as natural tasks for Speech-Audio Co-reasoning. In what follows, we explore the possibility of live music, specifically live music lessons, to be one such natural task.

3. StrumStart

In this section, we motivate the creation of StrumStart dataset and provide detail on how we collected and transcribed the audio data and question-answer pairs.

3.1. Speech-Music Co-reasoning (SMC)

Music lessons are a complex task that require participants to coordinate speech and audio in real-time [15, 16], and, for LALMs, offer an opportunity to test speech-audio co-reasoning capabilities. An example of a typical interaction, as shown in Figure 1, involves the student speaking and playing the guitar, and the teacher then responding in turn. To answer the question "Why does the teacher say "pretty good?", a model must make use of both the speech from the student, audio from the guitar, and understand their relationship with the teacher's response. Solely focusing on speech, in this example, results in an incorrect answer. Music lessons provide situation after situation where such speech-music events occur, and the possibility for further question generation is immense. To provide a testing ground for SMC, we collect the StrumStart dataset, as described below.

3.2. Data Overview

The StrumStart dataset is created from approximately 4 hours of guitar lessons between one teacher and one student, recorded on a Google Pixel 6 Android smartphone to mimic in-the-wild data collection conditions. 5 half hour to 1 hour-long lessons were recorded with one to two weeks between lessons. One 30 minute recording was also made where the teacher played various chords and notes for music-only questions. The student had no prior guitar experience and was expected to practice in-between lessons, which progress from basic guitar construction to beginner chords. The teacher is, by their own description, an intermediate guitarist who had never taught a student before. The student learns to play acoustic fingerstyle guitar while the teacher provides feedback on her own electric guitar. Lessons were a mix of hands-on playing and explanation of music theory concepts. A high-level overview of concepts covered and lesson duration are provided in Table 1. To increase data usability and encourage ethical dataset construction, no copyrighted music was performed during the lessons.

3.3. Transcription and Chunking

Each lesson's audio was transcribed with Whisper and 40-134 data points were generated per lesson, resulting in 580 samples in total. Each data point represents a clip of the audio that exhibits potential for a Speech-Music Co-reasoning question. This includes instances of speech related to what is being played on the guitar, question-answer dialogue between the teacher and student, corrections of erroneous playing during the lesson, and more. Pure clips of music are also isolated to test the models' ability to answer music-only queries. The clip's transcript was saved along with question about the audio and the answer to that question by a single human annotator.

Questions included identifying what was being played based on the dialogue, which parts of the guitar were being referred to, and why interruptions in speech and audio occurred. SMC questions were designed to gauge if the model could use both the music and the dialogue to answer questions about what is happening in the lesson. Music only questions were designed to test a model's ability to answer basic music questions such as what notes/chords were played, or what instrument was playing. Additionally, since this data was tailored to the application of a music lesson, some questions were designed to check if the model could understand if the student played correctly or improved and what the goal of the teacher was in certain instructions and speech.

Additionally, each data point was classified as a Speech-Only Reasoning, Speech-Music Co-reasoning, or Music-Only Reasoning task. Speech-Only Reasoning tasks were question-answer pairs where the question could be answered solely on the transcript. Speech-Music Co-reasoning tasks were labeled as such if they needed the information of guitar, pauses, or other audio to accurately answer the question. Music-Only Reasoning tasks involved the model answering specific questions about the music played (chords, notes, etc.) or identifying the instrument being used. About 53.1% of the questions were categorized under Speech-Only Reasoning, 31.2% questions were considered Speech-Music Co-reasoning, and 15.7% were labeled as Music-Only.

4. Model Evaluation

We utilize the question-answer (QA) pairs from StrumStart to evaluate five models: LTU (Listen, Think, Understand), LTU-AS, GAMA, QWEN, and SALMONN developed by [7], [2], [6], [8], and [1] respectively. All these models are LALMs that focus on general audio comprehension. LTU and LTU-AS are similar models, with the core difference being that LTU-AS was trained on speech tasks as well. The other models are also trained on general audio comprehension and reasoning. SALMONN and QWEN specifically note capabilities of music based reasoning, and GAMA is also trained in 'complex reasoning'.

Table 1: *Overview of Session Content and Duration*

Session	Content	Duration
Session 1	Guitar Basics, Fingering	35:37
Session 2	Triads, Chords	41:41
Session 3	Chord Progressions, Circle of Fifths	1:01:49
Session 4	Basic Chords, Spiderwalk	46:08
Session 5	Chord Practice, Strumming Patterns	47:23
Session 6	Chord and Note Demonstration (Teacher Only)	30:00

4.1. Methodology

For each QA pair in the dataset, the audio clip and associated question are passed to the models to answer the question. Those answers are saved, compared to the answers written via the human annotation, and evaluated through a 3 tier scoring system:

- 0 is assigned if the model outputs something completely incorrect, unrelated, or outputs that the question cannot be answered based on the audio.
- 0.5 is assigned if the model outputs an answer that is relevant to the question but still missing the correct answer or too vague
- 1 is assigned if all relevant information from the written answer is present in the output and no incorrect statements are made. Differences in wording are not penalized.

As per [7], existing benchmarking metrics, such as SPICE [17] are not optimal for language models trained on diverse datasets as synonymns are often failed to be considered. Since human evaluation is considered to be the gold standard in LALM evaluation [6], we utilize a 3 tier scoring system was created to serve as a metric for the models on the dataset.

4.2. Results

Overall, LTU-AS outperformed all the other models, with SALMONN following close behind. Still, both models' had significant percentages of incorrect answers for each lesson. Across all sessions LTU-AS scored 0 on 47.9% of data points, 0.5 on 16.1% data points, and 1 on 36.0% of data points. Across all sessions, SALMONN scored 0 on 64.0% of data points, 0.5 on 8.0% data points, and 1 on 28.0% of data points. The percent scores for every model on all the questions are displayed in Table 2.

Table 2: *Percent Scores For Each Model On All Questions*

Model	% Scored 0	% Scored 0.5	% Scored 1
LTU	90.4	3.6	6.0
LTU-AS	47.9	16.1	36.0
GAMA	90.4	3.6	6.0
QWEN	65.8	15.9	18.3
SALMONN	64.0	8.0	28.0

Per each lesson, the percentage of LTU-AS answers that were scored as 1 was never above 50%. The percentage of 0 and 1 scorings were even in each lesson, with 0.5 accounting for the rest of the performance. In terms of incorrect answers, GAMA and LTU often gave answers along the lines of "there is not enough information to answer the question." The other models rarely did so and provided confident, but incorrect answers.

For speech only questions, the models had mixed results in terms of performance. LTU-AS and SALMONN did generally well. They were able to answer some questions regarding the correctness of the student's answer given the teacher's questions and other basic comprehension tasks for the conversation. QWEN, LTU-AS, and SALMONN directly quoted the dialogue in some of its answers, which led to results that didn't answer the specific question asked in the prompt. For questions such as "Why does the teacher apologize?" or "Why does the teacher not want to focus on the technique mentioned?" examining the reasons for speech, LTU largely continues the trend of declaring there is not enough information, while the other models generate answers that may be logical reasons for the given question out of context, but still miss the exact reason in the audio.

Many speech-music questions arose from the context of the audio being a guitar lesson. In analyzing the models' responses to questions regarding why the teacher plays in the middle of speaking, or what the student is doing incorrectly while playing, we are able to measure their capabilities in SMC. Overall, the models did not perform as well on these questions. For example, in a clip where the teacher was speaking and then paused to play A minor in the middle to double check they were playing it correctly, none of the models could answer the question of "Why does the teacher pause in the middle of speaking and start playing?"

In terms of music only questions, all 5 models were largely unable to identify the notes and chords being played. While the other models attempted to identify the notes and did so incorrectly, GAMA would give an answer of "Without specific information about the chords being played, it's impossible to determine which exact chord the teacher is playing." Similarly, when the models were asked what technique was being used in the music-only recordings, the other models would generate incorrect answers while GAMA remained noncommittal. For music-only questions focused on identifying the instrument being played, all models were generally accurate in identifying that it was a guitar, but did erroneously answer that it was a ukelele or adjacent string instrument at times. For some of those questions, GAMA and QWEN answered with completely incorrect instruments such as piano and trumpet. Despite not being prompted, QWEN also attempted to identify the genre being played when asked "What instrument is being played in this clip?" However, as these recordings were taken in the context of introductory guitar lessons, there were no songs played and these genre classifications were inapplicable.

Some questions went beyond the realm of music and focused more on interpersonal conversation. For example, for a portion of the audio where the teacher had a prolonged silence after the student asked a question and then apologized, a question was posed: ”Is it polite of the teacher to pause for so long after hearing the student’s question?”. The annotator’s answer was that it was impolite of the teacher to do so. LTU-AS came to a similar conclusion: ”It is not necessarily impolite for a teacher to pause for so long after hearing a student’s question, but it could be seen as rude if they are intentionally ignoring or dismissing the student’s request for help or attention.” This shows an understanding of conversational etiquette and the ability to reason as to why the action may be taken as impolite.

Focusing on LTU-AS’ performance, there is a difference in performance between tasks labeled as Speech-Only, Speech-Music Co-reasoning, and Music-Only, as shown in Figure 2. LTU-AS had majority correct (score of 1) answers for tasks that would only need the transcript to be answered. For speech-music questions, LTU-AS performed worse, but still answered a significant amount of questions correctly. For music-only questions, LTU-AS answered overwhelmingly incorrectly with about 70% of answers being scored 0. This is likely due to the large number of incorrect answers for chord, note and technique identification tasks. As mentioned, LTU-AS was successful in identifying the instrument played which contributes to the proportion of music-only questions with scores of 1.

SALMONN performed better on music only questions, but much worse on Speech-Music Coreasoning, as shown in Figure 3. This is likely because SALMONN got all the instrument identification questions correct. It is clear that while LTU-AS and SALMONN had similar overall accuracies, their specific abilities in speech-only reasoning, SMC, and music-only reasoning differ.

This performance demonstrates that SMC and Music Reasoning are difficult tasks requiring further exploration and modeling efforts.

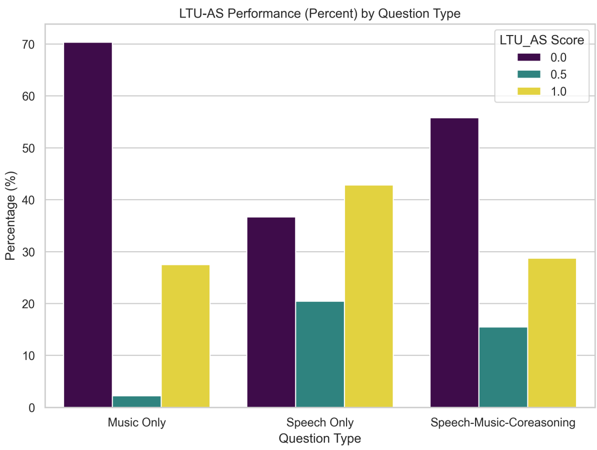

Figure 2: *Percentages of 0, 0.5, and 1 scores split by question type for LTU-AS*

5. Discussion and Future Work

Despite large room for improvement, SALMONN and LTU-AS’ performance on the unseen StrumStart dataset are a promising development. The decreased performance of these models across increasing levels of task complexity suggests that

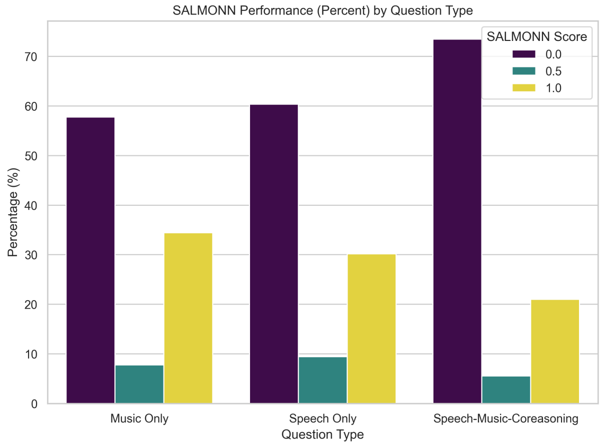

Figure 3: *Percentages of 0, 0.5, and 1 scores split by question type for SALMONN*

Speech-Music Co-reasoning and Music reasoning are worthy tasks to refine. In future work, we plan to explore finetuning these models to this task and generate more data.

Although used in this work as an evaluation dataset, there are multiple avenues to increase StrumStart’s potential as a training set for Speech-Music Co-reasoning. Additional lessons that focus on more difficult topics as the student’s skill increase would provide additional data. Furthermore, in the existing lessons, more manual extraction of QA pairs can be conducted to collect more Speech-Music Co-reasoning examples. Since the potential for new Speech-Music Co-reasoning QA pairs is quite high, StrumStart is highly suited for the sort of synthetic data generation seen applied to other datasets.

Finally, an interesting consequence of StrumStart is the question of expertise. As the teacher themselves was an amateur guitarist, another avenue of expansion would be for an expert or experienced guitar teacher listen provide their own QA pairs, possibly noting flaws in the StrumStart teacher’s pedagogy. Seen from the perspectives of student, teacher, and observer, the complexity of live music and music teaching provide a seemingly limitless possibility for the creation of complex and realistic multi-modal co-reasoning tasks.

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